

Margaret Ma

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EDUCATION

University of Waterloo

Candidate for Bachelor of Applied Science, Honours Electrical Engineering

Waterloo, ON

Sep. 2023 – Present

- Relevant Courses: Electronic Circuits 1, Digital Computers, Signals & Systems

PROJECTS

Quadruped Robot Hardware Platform | *Mechatronics, Hardware Integration*

May 2026 – Present

- Designed and integrated the electrical architecture of an **8-DOF quadruped robot** using **Raspberry Pi 4, LX-16A serial bus servos**, buck converters, and distributed power electronics
- Developed Python software on **Raspberry Pi OS** for **servo communication**, actuator control, and hardware integration
- Implemented **forward and inverse kinematics** in **MATLAB** to model leg motion and generate joint trajectories for autonomous locomotion
- Manufactured and assembled the robot chassis using **FDM 3D printing (PLA)**
- Performed **hardware bring-up**, subsystem testing, and debugging of embedded controllers, communication buses, and power systems

Autonomous Quadcopter | *ArduPilot, Pixhawk, BLDC Motors, Schematic*

Nov 2025 – Present

- Designed the electrical architecture for a **Pixhawk**-based autonomous drone integrating **Pixhawk, Raspberry Pi, GPS, RC receiver, ESCs, BLDC motors**, telemetry, and LiPo power distribution
- Configured **ArduPilot** firmware, **Mission Planner**, radio calibration, ESC setup, and flight controller parameters for hardware validation and testing
- Configured **Raspberry Pi OS** on a **Raspberry Pi 4** companion computer for future autonomous flight development and embedded robotics applications

STM32 Minimum System PCB | *Circuit Design, PCB Layout, STM32CubeIDE*

July 2025 – August 2025

- Designed a minimal **STM32** system PCB including **power regulation**, decoupling capacitors, and clock/reset circuitry for stable MCU initialization
- Validated **USART, I²C, and USB** peripheral interfaces in **STM32CubeIDE** through bring-up testing and debugging

EXPERIENCE

Hardware Engineering Intern, WATbotics

May 2026 – Aug 2026

University of Waterloo

Waterloo, ON

- Designed **PCB schematics and layouts**, and selected electronic components for autonomous robotic systems
- Evaluated **servo motors**, communication protocols, and power electronics for robotic platforms
- Configured **Raspberry Pi** environments, documented setup procedures, and managed electronic component procurement

Electrical Quality Technician, Gordie Howe International Bridge

Sep 2024 – Dec 2024

Fluor Corporation

Windsor, ON

- Conducted inspections of **UPS systems, transformers, grounding networks, and elevator electrical rooms** on large-scale infrastructure, ensuring compliance with engineering ITPs and test plans
- Interpreted **electrical specifications and quality requirements** to verify compliance documentation against engineering standards

TECHNICAL SKILLS

Programming Languages: C++, Python, RISC-V Assembly, Matlab

Embedded Systems & Robotics: STM32, STM32CubeIDE, Raspberry Pi 4, Raspberry Pi OS, Pixhawk, ArduPilot, UART, SPI, I²C, Linux, Git

PCB Design & Hardware Development: PCB Design, Schematic Capture, KiCad, Altium Designer, LTspice, Circuit Debugging, Embedded Systems, Power Electronics, Sensor & Actuator Integration

Mechanical Design & Manufacturing: SolidWorks, Autodesk Fusion 360, AutoCAD, FDM 3D Printing

Prototyping & Test Equipment: Oscilloscope, Digital Multimeter, Function Generator, Power Supply, Soldering